YEAR 12 GENERAL MATHEMATICS INVESTIGATIVE TASK 3

Sequences and series

Question 1

A new fitness centre has a number of different programs aimed at increasing strength and cardio fitness. One 12-week program offered focuses on improving running stamina. In the first week, participants must run 4 km, and each week afterwards the number of kilometres they must run increases by two.

a Write an equation to describe this situation.

1 mark

b How many kilometres will participants run in their fifth week of the program?

1 mark

c How many kilometres will participants run in the entirety of the program?

1 mark

The success of the program prompts the organisers to extend it for the rest of the year.

d After how many weeks will the participants be running 50 km?

1 mark

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e After how many weeks will the participants have covered 1000 km?

1 mark

Another program looks at increasing the amount of weight people can lift. The program promises it will improve strength by 80% in the first month, and then 50% of that each month thereafter. One particular participant can lift 50 kg before starting the program.

f How much weight will the participant be able to lift after the:

i first month

ii second month?

2 marks

g Write an equation to describe the cumulative improvement in weight-lifting ability.

1 mark

h How much will this participant be able to lift after 4 months?

2 marks

i What is the maximum weight that this person will be able to lift?

1 mark

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Enrolments for each program are increasing. The current enrolment for the running program is 35 people, which increases by 8% each term. (There are four terms per year.)

j What is the number of expected enrolments in the running program after 2 years?

1 mark

k Plot a graph of the enrolments in the running program over the 2 years.

2 marks

l If R_n represents the enrolment in the running program after n terms, write a difference equation in terms of R_n and R_{n+1}.

1 mark

m In which term will the enrolment in the running program first exceed 100 people?

1 mark

Enrolments for the strength program can be described by the difference equation:

$$W_{n+1} = 1.15W_n - 3$$

where n represents the number of terms. The initial enrolment for the strength program is 28.

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n Plot the first 2 years of this sequence on the same set of axes used in part **k** and, hence, find the expected enrolment in the strength program after 2 years.

3 marks

o After how many years will the enrolments in the strength program be greater than the enrolments in the running program?

2 marks

Total 21 marks

Question 2

In a particular city, two recycling plants open at the start of the year. The amount of material each plant recycles for each of 3 months is shown in the table below.

	Material recycled (tonnes)	
Month	Plant A	Plant B
1	6	2
2	7.5	2.4
3	9	2.88

a For each recycling plant, write a rule for t_n, the amount of material recycled in the nth month, expressed in terms of n.

2 marks

b Complete the table by finding how much material each plant recycles in the fourth month of the year.

1 mark

c How much material will each plant recycle in December?

2 marks

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d Find the total amount of material recycled by each plant in the first 2 years.

2 marks

e In which month does Plant B overtake Plant A in the amount of material recycled?

2 marks

Once a recycling plant is recycling 100 tonnes of material per month, it needs to expand to accommodate the higher volumes.

f How many months earlier will Plant B need to undertake renovations than Plant A?

3 marks

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After every 1000 tonnes that a plant recycles, it needs to undergo inspections to ensure that all equipment is in good order.

g How many inspections will Plant B undergo before Plant A has its first inspection?

4 marks

Total 16 marks

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Question 3

The Mountain pygmy possum is Australia's only hibernating marsupial. Recent estimates put the wild population at less than 2000 individuals. Healesville Sanctuary in Victoria has a captive breeding population that is currently at 45 individuals. This population increases by approximately 15% each year.

a Write a rule for t_n to describe this situation.

1 mark

b Use your rule to find the population of the possums after 10 years.

1 mark

When the population reaches 200, it is planned to establish a new breeding program in NSW with 50 of these possums.

c After how many years will this new breeding program be established?

2 marks

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This new breeding program finds that it is not as successful as the original breeding program, with the population increasing by roughly 9% every year. In order to boost the numbers, Healesville gives NSW an extra four possums at the end of each year.

d Write a difference equation for both the new breeding program and that at Healesville Sanctuary.

2 marks

e Use one of these difference equations to find the population of the NSW breeding program in 5 years' time.

2 marks

f When will the new breeding program reach 200 individuals?

1 mark

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g When the combined population from both programs reaches 1000 individuals, they will release 200 possums into the wild. When will this occur?

3 marks

Total 12 marks

Question 4

According to an old legend, the king of Iran heard that somebody had invented a game called chess. He asked the inventor to teach him, and to choose any reward for doing so. The inventor simply asked for a number of grains of rice according to the following rule.

- One grain for the first square on the chess board.
- Two grains for the second square.
- Four grains for the third square.
- Eight grains for the fourth square and so on.

a Write a rule for t_n, the number of grains of rice on the nth chess square, expressed in terms of n.

1 mark

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b Hence, find the number of grains on the 20th square.

1 mark

c If 1 kg of rice contains approximately 50 000 grains, find the total mass of rice the inventor has requested as payment.

2 marks

The king quickly realised that he would be unable to give this amount of rice as payment. A potential counter offer was 1 000 000 000 kg of rice for the first square, then half this amount for the second square, then half this again for the third square and so on.

d Write a rule for t_n, the weight of rice on the nth chess square, expressed in terms of n.

1 mark

e Using this sequence rather than the original sequence, how many more grains of rice are on the 20th square?

3 marks

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f Show which method of payment is greater.

3 marks
Total 11 marks

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